

MIET AFRICA

Educational Technology Platform

Web Application & Mobile App

PHASE 1 — REQUIREMENTS DOCUMENT

Client: MIET Africa NGO, South Africa
Date: February 2026
Document Version: 1.0

Prepared for school project submission — GitHub Repository Upload

1. Client Profile

This section provides a detailed background of our selected NGO client, MIET Africa, including their mission, the communities they serve, and the real-world problems that our technology solution will address.

Field	Details
Organisation Name	MIET Africa
Type	Non-Governmental Organisation (NPO/NGO)
Founded	1990
Headquarters	Durban, KwaZulu-Natal, South Africa
Website	www.mietafrica.com
Primary Focus	Education & Youth Development
Target Beneficiaries	Children and youth in vulnerable, under-resourced school communities
Geographic Reach	South Africa and sub-Saharan Africa
Key Partners	Government departments, schools, UNICEF, and international donors
Languages Served	English, Zulu, Xhosa, Afrikaans, and Sotho

1.1 Organisation Background

MIET Africa is a South African NGO that has been working since 1990 to improve the quality of education for children and youth living in vulnerable and impoverished communities. The organisation implements comprehensive education programmes that go beyond simply providing resources. They address the root causes that keep children from learning effectively, including poverty, lack of trained teachers, language barriers, and social inequality.

MIET Africa works directly with schools, educators, parents, and local communities. Their approach is holistic: they understand that a hungry, unsafe, or emotionally unsupported child cannot learn effectively, so their programmes account for the full context of a child's life. The organisation has impacted thousands of learners across South Africa, with a particular focus on KwaZulu-Natal, Limpopo, and Eastern Cape — provinces with some of the highest rates of educational disadvantage in the country.

1.2 The Real-World Problem

Despite MIET Africa's dedicated efforts, significant operational and technological gaps prevent them from reaching their full potential. The following problems are what our digital platform will directly solve:

- **Isolation of rural educators:** Teachers in remote township and rural schools have no access to digital professional development resources, leaving them without ongoing training or support.
- **Manual paper-based tracking:** Student progress is recorded on paper, making it impossible for the NGO to identify struggling learners at scale or measure programme impact accurately.
- **Communication breakdown:** There is no efficient communication channel between MIET Africa headquarters and partner schools, leading to delayed responses and missed opportunities for intervention.
- **No centralised monitoring system:** MIET Africa cannot monitor what is happening across hundreds of schools simultaneously, making strategic planning and donor reporting very difficult.
- **Language exclusion:** A large portion of children and educators cannot access resources in their home language, reducing comprehension and engagement with learning materials.
- **Donor transparency gap:** Donors and funders have no real-time visibility into where funds are going or what measurable impact they are achieving.

1.3 Why MIET Africa is the Ideal Client

MIET Africa is the ideal NGO client for this project because their work directly addresses one of South Africa's most urgent social crises — educational inequality. A digital platform for MIET Africa is both technically feasible for a student project and capable of creating genuine, measurable realworld impact. Both a web app (for admin and monitoring) and a mobile app (for teachers and students) are not just useful — they are essential to solving the problems identified above.

2. Project Overview

2.1 Solution Summary

We will design and build a dual-platform digital solution for MIET Africa consisting of two interconnected applications:

- **Mobile Application:** Designed for teachers, students (senior grades), and community volunteers. Focuses on learning resources, student progress tracking, literacy assessments, and teacher training — all available offline.
- **Web Application:** Designed for MIET Africa administrators and partner school principals. Focuses on monitoring, reporting, resource management, school registration, and donor impact reporting.

Together, these two platforms create a connected education ecosystem. Every action taken on the mobile app feeds data into the web dashboard, giving MIET Africa real-time visibility into what is happening across all their partner schools.

2.2 How This Technology Helps Children

Every feature built into this system exists for one ultimate purpose: improving educational outcomes for vulnerable South African children. The table below directly maps how our technology helps children:

Child's Real Problem	How Our App Solves It	Direct Benefit to the Child
Teacher lacks skills and training	Teacher Training Modules on the mobile app	Child receives better, more confident teaching every day
No learning materials at school	Offline Resource Library downloadable on mobile	Child accesses quality lessons even without internet
Falling behind undetected	Literacy Assessment Tool with instant results	Struggling learners identified early and given targeted help
Language barrier blocks learning	Multilingual support: Zulu, Xhosa, Afrikaans, English	Child understands content in the language they speak at home
Teacher isolated from NGO support	Push notifications and in-app messaging system	Teacher gets timely help; child's classroom environment improves
School needs go unrecognised	School profiling and needs assessments via the app	MIET Africa sends targeted resources to that specific school
Child's growth is never measured	Student progress tracker with visual analytics	Child's improvement is tracked, celebrated, and acted upon

3. Functional Requirements

Functional requirements describe what the system must DO — the specific features and behaviors it must deliver to its users. Each requirement is labelled with a unique ID for traceability on GitHub.

3.1 Mobile Application — Functional Requirements

FR-M01: User Registration and Authentication

- Teachers and senior students must register with their name, school name, province, and role.
- Users must log in securely using email and password.
- Password reset functionality must be available via email or SMS.
- Session tokens must be stored securely on the device.
- **Child Impact:** Verified educators mean only trained, trusted adults are guiding learning through the platform.

FR-M02: Offline Learning Resource Library

- The app must display a searchable library of lesson plans, worksheets, educational videos, and teaching guides.
- Resources must be filterable by grade (Grade 1 to 12), subject, and language.
- Teachers must be able to download resources for full offline access.
- New resources added by MIET Africa must sync automatically when the app reconnects to the internet.
- **Child Impact:** Children in schools with no reliable internet can still access quality, curriculum-aligned learning materials.

FR-M03: Student Progress Tracker

- Teachers must be able to create a digital class roster with individual student profiles.
- Teachers must be able to log reading levels, assessment scores, and learning milestones per student.
- The app must display a visual progress graph per student over time.
- Progress data must sync to the web dashboard when internet is available.
- **Child Impact:** No child falls through the cracks — struggling learners are identified and supported before they fall too far behind.

FR-M04: Literacy Assessment Tool

- Teachers must be guided through structured literacy assessments using a step-by-step question interface.
- The system must generate an instant report categorising the student as: Below Grade Level, Approaching Grade Level, On Track, or Excelling.
- Assessment history must be stored per student and visible over time.
- Results must sync to the web admin dashboard when connectivity is available.
- **Child Impact:** Standardised assessments ensure every child is evaluated fairly and no struggling learner is overlooked.

FR-M05: Teacher Professional Development Modules

- The app must host short video lessons, reading guides, and knowledge quizzes for teacher development.
- Teachers must be able to track their own training completion with a visual progress indicator.
- Certificates of module completion must be generated and shareable as PDF.
- All modules must be available offline once downloaded.
- **Child Impact:** A better-trained, more confident teacher delivers higher quality lessons — children benefit directly.

FR-M06: Push Notifications and Announcements

- MIET Africa staff must be able to send targeted push notifications to specific schools, provinces, or all users.
- Notification types must include: new resource alerts, upcoming training events, urgent announcements, and motivational messages.
- Users must be able to manage their notification preferences in their settings.
- **Child Impact:** Teachers remain informed, motivated, and connected — a supported teacher creates a better learning environment for every child.

FR-M07: Multilingual Interface and Content

- The full app interface must be available in English, isiZulu, isiXhosa, and Afrikaans.
- Educational resources must be tagged by language and filterable accordingly.
- Users must be able to set and change their preferred language in their profile settings.
- **Child Impact:** Children in home-language instruction schools engage better with materials they can actually understand.

FR-M08: Offline Mode and Automatic Data Sync

- All downloaded resources, assessment tools, and student data must function without an internet connection.
- The app must clearly indicate which content is available offline.
- When connectivity is restored, all locally stored data must sync automatically to the server.
- **Child Impact:** Rural and township schools — which suffer most from poor connectivity — are not excluded from any platform benefits.

3.2 Web Application — Functional Requirements

FR-W01: Admin Dashboard and Analytics

- MIET Africa staff must see a real-time overview of: total registered schools, total teachers, student assessments completed, and resource downloads.
- The dashboard must include visual charts: literacy improvement rates by province, training completion rates, and resource usage trends.
- All data must be exportable to CSV and PDF formats for external reporting.
- **Child Impact:** MIET Africa can see exactly which schools need urgent help and deploy resources where they matter most for children.

FR-W02: Resource Upload and Content Management

- Administrators must upload new lesson plans, videos, and guides with metadata including grade, subject, language, and tags.
- A content approval workflow must allow senior staff to review and approve materials before they are published to teachers.
- Administrators must be able to edit, archive, or remove resources from the library.
- **Child Impact:** Only quality-checked, curriculum-aligned content reaches teachers and children — no misinformation or substandard materials.

FR-W03: School and Teacher Registration Portal

- Schools must submit a registration request with their profile, GPS location, province, and number of enrolled learners.
- MIET Africa admins must be able to approve, reject, or flag school registrations with comments.
- Admins must view a full directory of all registered schools with filtering by province, status, and size.
- **Child Impact:** A verified network of schools ensures MIET Africa's tools reach real, accountable educators rather than unknown users.

FR-W04: Reporting and Impact Tracker

- The system must generate monthly and quarterly impact reports automatically based on captured data.
- Reports must show: number of children assessed, literacy improvement percentages, training hours logged by teachers, and total resources accessed.
- Reports must be downloadable in PDF format for donor and government submissions.
- **Child Impact:** Accurate reporting secures donor funding, which directly funds the programmes that benefit children on the ground.

FR-W05: Event and Training Scheduler

- MIET Africa staff must be able to create training workshops with date, location, topic, and maximum attendee count.
- Teachers must be able to register for events via both the mobile and web applications.
- Automated reminder push notifications must be sent 48 hours before each event.
- **Child Impact:** Better-attended teacher training events mean more children benefit from skilled and motivated educators.

FR-W06: Donor and Volunteer Portal

- A public-facing section of the web app must display key impact statistics in a visually engaging way.
- Donors must be able to view which schools are being supported, what programmes are running, and live impact numbers.
- Volunteers must be able to submit an expression of interest form from the website.
- **Child Impact:** Sustained donor support funds the NGO's operations, keeping all childfacing programmes running and growing.

4. Non-Functional Requirements

Non-functional requirements describe HOW the system must perform — the quality attributes and constraints that ensure the system is reliable, safe, usable, and legally compliant.

ID	Category	Requirement	Why It Matters for Children
NFR01	Performance	The mobile app must load the home screen within 3 seconds on a standard Android device.	Teachers in rural schools use lowend phones. Slow apps get abandoned and children lose access.
NFR02	Offline Capability	The app must function fully offline for all core features after initial setup and download.	Many target schools have no reliable internet. Offline mode ensures no child is left out.
NFR03	Security	All user data must be encrypted in transit (HTTPS/TLS) and at rest (AES-256).	Student data is sensitive and legally protected. Children's privacy must be safeguarded.
NFR04	Scalability	The system must support up to 10,000 concurrent users without performance degradation.	MIET Africa works across hundreds of schools and needs the system to grow with their reach.
NFR05	Usability	The mobile app UI must be usable by a teacher with no prior app experience within 10 minutes.	Many educators in target communities have limited smartphone experience.
NFR06	Accessibility	Text must meet WCAG 2.1 AA contrast standards. Font sizes must be adjustable by the user.	Inclusive design ensures no educator or learner is excluded due to visual limitations.
NFR07	Compatibility	Mobile app must support Android 8.0 and above. Web app must support Chrome, Firefox, and Safari.	Most affordable Android phones in South Africa run Android 8 or higher.
NFR08	Availability	The web server must maintain 99.5% uptime with automated failover.	NGO staff rely on the web dashboard for daily operations and donor reporting.
NFR09	Data Privacy	The system must comply with South Africa's POPIA (Protection of Personal Information Act).	Legal compliance is mandatory when collecting and storing children's educational data.
NFR10	Maintainability	The codebase must follow documented coding standards with at least 70% unit test coverage.	Enables future developers at MIET Africa or student teams to maintain and extend the system.

5. User Requirements

User requirements are written from the perspective of each type of user — what they need the system to do for them, in plain non-technical language. These were gathered through background research and will be validated during the client interview.

5.1 Teacher (Primary Mobile App User)

- As a teacher, I want to download lesson plans so that I can use them in class even when there is no internet connection.
- As a teacher, I want to track each student's reading level so that I can see clearly who is falling behind.
- As a teacher, I want to complete training modules on my phone so that I can develop my skills in my own time without travelling.
- As a teacher, I want to receive notifications about new resources so that I always have the latest teaching materials available.
- As a teacher, I want to conduct literacy assessments and get instant results so that I know exactly where each child stands academically.

5.2 MIET Africa Administrator (Primary Web App User)

- As an admin, I want to see a dashboard of all registered schools so that I can monitor which areas need the most support.
- As an admin, I want to upload and approve educational resources so that teachers always have access to quality content.
- As an admin, I want to generate impact reports so that I can submit accurate programme data to our donors on time.
- As an admin, I want to schedule training events and track registrations so that workshops are well-organised and well-attended.
- As an admin, I want to approve new school registrations so that only verified, legitimate schools join the platform.

5.3 School Principal (Secondary Web App User)

- As a principal, I want to see my school's literacy progress data so that I can report meaningfully to parents and the education department.
- As a principal, I want to register my school on the platform so that my teachers can immediately access MIET Africa resources.
- As a principal, I want to see which teachers have completed their training modules so that I can encourage those who have not yet started.

5.4 Donor / Funder (Web App Read-Only User)

- As a donor, I want to see how many children have been helped this year so that I can evaluate whether to continue or increase my funding.
- As a donor, I want to view and download impact reports so that I can include them in my own organisational reporting and accountability documents.

6. Proposed System Architecture

This section provides a high-level overview of the technologies that will be used to build the platform. The full technical architecture, including entity relationship diagrams and system diagrams, will be produced in Phase 2.

System Layer	Proposed Technology	Purpose
Mobile App Frontend	SwiftUI	Cross-platform mobile app for both Android and iOS from a single codebase
Web App Frontend	React.js	Admin dashboard, school portal, and public donor pages
Backend API	Node.js with Express	RESTful API handling all data requests from both applications
Database	PostgreSQL	Stores all user, school, student, and resource data securely
File Storage	AWS S3 or Cloudinary	Stores uploaded educational resources including PDFs and videos
Authentication	JWT (JSON Web Tokens)	Secure, stateless user authentication across both platforms
Offline Storage	SQLite on device	Local storage for offline mode; automatically syncs when online
Push Notifications	Firebase Cloud Messaging	Delivers push notifications to all registered mobile devices
Cloud Hosting	Vercel	Reliable, scalable cloud hosting for the web application and API
Version Control	GitHub	Code repository, project documentation, and team collaboration

7. Client Interview Questions

The following questions must be asked during the Phase 1 client interview with a representative from MIET Africa. The answers will validate and refine the requirements documented in this document before development begins.

7.1 Operational Questions

- How many schools and teachers are currently part of MIET Africa's active programmes?
- How do you currently distribute educational materials to partner schools?
- How is student progress currently tracked across your partner schools?
- What communication channels do you use to stay in touch with teachers in remote areas?
- How often do teachers attend in-person training workshops, and what barriers prevent more teachers from attending?

7.2 Technology Questions

- What devices do most of your partner teachers use — Android, iPhone, or basic feature phones?
- What is the average internet connectivity situation at your partner schools?
- Does MIET Africa have any existing digital tools or systems we need to integrate with or replace?
- Who on your team would manage and maintain the web dashboard daily?

7.3 Impact and Reporting Questions

- What key metrics do your donors require in your impact reports?
- How do you currently measure literacy improvement across your programmes?
- What would success look like for this technology platform after 12 months of use?
- Are there any legal or data privacy requirements specific to your funders that we should know about?

8. Project Deliverables Timeline

The table below outlines the key deliverables for the full project, starting from Phase 1. This document represents the Phase 1 submission.

Phase	Deliverable	Target Deadline
Phase 1	Requirements Document (this document) uploaded to GitHub repository	Week 3
Phase 1	Client Profile and Interview Notes documented	Week 3
Phase 2	System Design: ERD, Use Case Diagrams, and App Wireframes	Week 5
Phase 3	Working Prototype — Mobile App with core features	Week 8
Phase 3	Working Prototype — Web App admin dashboard	Week 8
Phase 4	Fully Tested Application: unit testing, integration, and UAT	Week 11
Phase 5	Final Presentation and complete GitHub Repository submission	Week 13

9. Document Sign-Off

This requirements document must be reviewed and signed off by all stakeholders before development proceeds to Phase 2.

Role	Name	Signature	Date
Student Author / Developer			February 2026
Project Supervisor / Lecturer			February 2026
Client Representative	MIET Africa		February 2026

End of Requirements Document — Version 1.0

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